



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year: 2023-2024 (Odd Semester)
Innovative Practices Description for Unit-I

Degree, Semester & Branch: III Semester B.E. AI&DS A Section
Course Code & Title: CS3351 Digital Principles and Computer Organization
Name of the Faculty member: Mr.B.Kannan

Name of the Topic	Name of the Innovative Practice	Date & Duration
Magnitude Comparator	One Minute Paper	29.09.2023 (10 Minutes)

Description:

- A "one-minute paper" may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question.
- It prompts students to reflect on the day's lesson and provides the instructor with useful feedback.

Goals (Learning Outcomes):

- The students will be able to describe the function of manganite comparator.

Use of Appropriate Method:

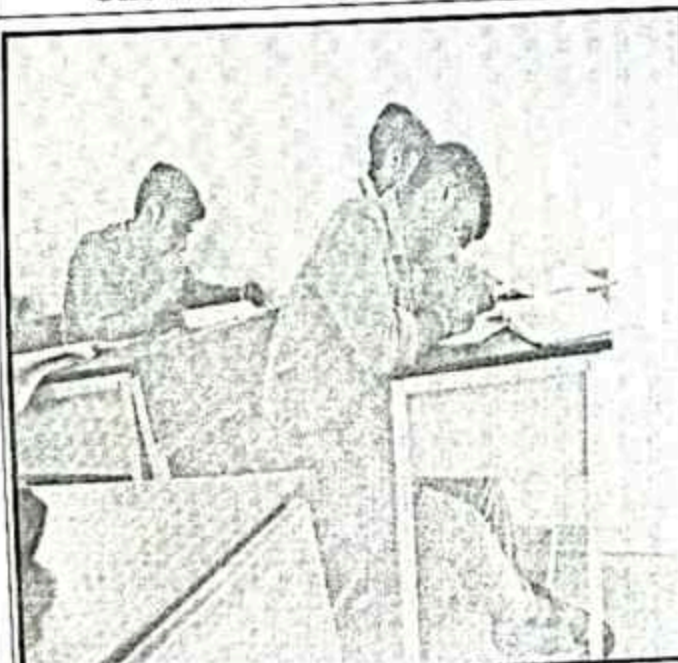
Justification for choosing the following Activities:

- The great advantage of minute papers is that they provide manageable amounts of timely and useful feedback for a minimal investment of time and energy.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

- Provide a "conceptual bridge" between successive class periods.
- Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

One Minute Paper for Manganite Comparator on 27.09.2023 (10 Minutes)



Output: $102 = A \oplus A \oplus A \oplus B \oplus B \oplus B$
Truth Table:

Input				Output		
A ₁	A ₀	B ₁	B ₀	A>B	A=B	A<B
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

K-Map for A>B:

AB	00	01	11	10
0	0	0	0	0
1	1	1	1	1
1	1	1	1	1

$$A > B = A_0 \bar{B}_0 + A_1 \bar{B}_1 + A_1 \bar{B}_0 + A_0 \bar{B}_1$$

Competency and COs	POs and PSOs												PSO1	PSO2	PSO3
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12			
C202.1	M	M	H	M	L	-	-	-	-	-	-	L	L	L	-

CO1. The students will be able to describe the magnitude comparator design and implementation.

References:

- ❖ <https://www.geeksforgeeks.org/magnitude-comparator-in-digital-logic/>
- ❖ M Morris Mano & Michael D.Ciletti, "Digital Design: With an Introduction to the Verilog HDL, VHDL, and System Verilog", Edition: Sixth, ISBN: 978-9353062019, Publisher: Pearson Education and Year: 2018.
- ❖ <https://de-iiit.vlabs.ac.in/exp/comparator-using-logic-gates/theory.html>

10/12/23
Faculty Member


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Innovative Practices Description for Unit-II

Degree, Semester & Branch: III Semester B.E. AI&DS A Section
Course Code & Title: CS3351 Digital Principles and Computer Organization
Name of the Faculty member: Mr.B.Kannan

Name of the Topic	Name of the Innovative Practice	Date & Duration
Flip Flops (SR & D) Operation, Excitation Table	Mind Map	17.10.2023 (10 Minutes)

Description:

- Mind maps activity enable quality engineering teams to construct a visual framework to aid in analysis and memory by providing a simple approach to generate concepts and ideas. They are often built around a single notion, rendered as a picture, and accompanied by representations of related concepts and relationships.

Goals (Learning Outcomes):

- The students will be able to understand the operation, excitation table and construction of flip-flops graphically.

Use of Appropriate Method:

Justification for choosing the following Activities:

- It is a graphical way to represent large information into a useful knowledge base and also helps in connecting the knowledge base with real time scenario. This facilitates the students to build better and new ideas.
- Mind map is a useful pictorial thinking tool, which is simple and effective. They show the relationship and hierarchy between ideas through visuals.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

- The mind map teaching method helps students organize concepts rapidly for abstract courses with high memory requirement.
- Students will use their learned topic to complete and extend their mind map task.

One Minute Paper for Manganite Comparator on 27.09.2023 (10 Minutes)

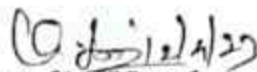


Competency and COs	POs and PSOs												PSO1	PSO2	PSO3
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12			
C202.3	H	H	M	M	H	-	-	-	-	M	L	L	L	L	L

CO1: The students will be able to design sequential circuits and analyze the design procedures.

References:

- ❖ <https://www.electronicsforu.com/technology-trends/learn-electronics/flip-flop-rs-jk-t-d>
- ❖ M.Morris Mano & Michael D.Ciletti, "Digital Design: With an Introduction to the Verilog HDL, VHDL, and System Verilog", Edition: Sixth, ISBN: 978-9353062019, Publisher: Pearson Education and Year: 2018.
- ❖ <https://de-iitr.vlabs.ac.in/exp/truth-tables-flip-flops/theory.html>
- ❖ <https://www.javatpoint.com/d-flip-flop-in-digital-electronics>


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Innovative Practices Description for Unit-III

Degree, Semester & Branch: III Semester B.E. AI&DS A Section

Course Code & Title: CS3351 Digital Principles and Computer Organization

Name of the Faculty member: Mr.B.Kannan

Name of the Topic	Name of the Innovative Practice	Date & Duration
Von Neumann Architecture & Operations and Operands	Write Pair Share	01.11.2023 (10 Minutes)

Description:

- One extension of think-pair-share is write-pair-share, in which students are given a chance to write down their answer before discussing it with their neighbor.
- Students think about the question on their own for a short period of time (usually 1-3 minutes). Pair. Students partner up with someone seated in close proximity (or small groups online) to discuss their solution to their answer and ask questions of their partner.

Goals (Learning Outcomes):

- The students will be able to understand the operation, excitation table and construction of flip-flops graphically.

Use of Appropriate Method:

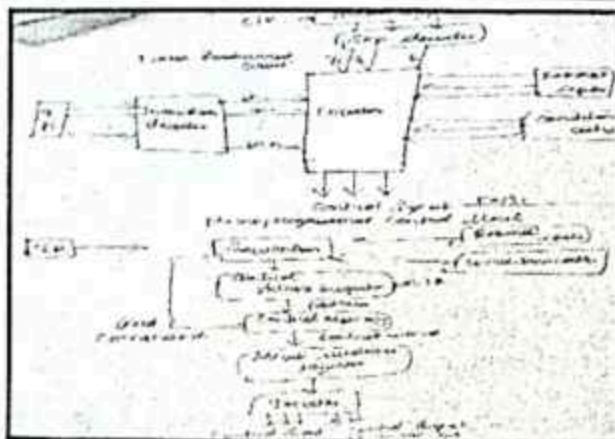
Justification for choosing the following Activities:

- TPS allows students to express their reasoning, reflect on their understanding and obtain prompt feedback on their learning.
- Write pair share activities lead to increased conceptual understanding and application of the topic.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

- Think Students: Write down the answer to the given question.
- Pair Students: (i) Identify parts of the answer that they have missed out.
(ii) Discuss their answers; do pros-cons analysis if there are multiple solutions.
- Share Instructor discusses: (i) What are all the essential parts in the answer? (ii) Pros-cons of various solutions given by students

Write Pair Share on Von Neumann Architecture & Operations and Operands on 01.11.2023 (10 Minutes)



Competency and COs	POs and PSOs												PSO1	PSO2	PSO3
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	L	L	-
C202.3	H	H	M	M	-	-	-	-	-	M	-	-	L	L	-

CO1: The students will be able to design sequential circuits and analyze the design procedures.

References:

- ❖ <https://www.electronicsforu.com/technology-trends/learn-electronics/flip-flop-rs-jk-t-d>
- ❖ M.Morris Mano & Michael D.Ciletti, "Digital Design: With an Introduction to the Verilog HDL, VHDL, and System Verilog", Edition: Sixth, ISBN: 978-9353062019, Publisher: Pearson Education and Year: 2018.
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- ❖ <https://www.javatpoint.com/d-flip-flop-in-digital-electronics>

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